Two forbidden patterns bound the boundary intersections of two simple heptagons

A computer-assisted proof that $f(7,7) = 38$,
modulo one imported general-position reduction

Jayden Lin*

17 September 2026

Abstract

Let $f(m,n)$ be the maximum number of points at which the boundaries of two simple polygons with m and n sides can meet. Dillencourt, Mount and Saalfeld posed the problem in 1993 and settled the cases where m or n is even; the odd–odd cases have resisted. We give a computer-assisted proof that $f(7,7) = 38$, conditional on one imported statement.

The mathematical content is a reduction. Rather than enumerate the 13,003,620 candidate crossing patterns with 40 intersections, we exhibit *two* local patterns, involving 8 and 14 prescribed side-pair incidences, and show that every candidate pattern with 40 intersections and every one with 42 contains one of them. Each of the two is non-realizable, and a degree argument over 28 cases reduces the whole exclusion to those two facts.

Everything in this paper except one statement is verified by the Lean 4 kernel from primitive definitions, with no appeal to any published lemma. The exception is the reduction to general position, for which we import Ackerman–Keszegh–Rote §9 and carry it as an explicit hypothesis. We are explicit throughout about which assertions are machine-checked, which are imported, and which — notably the faithfulness of our definitions — remain matters for a human reader.

Contents

1 Introduction **2**
1.1 The problem . . . 2
1.2 What is proved here . . . 2
1.3 What is new . . . 3

2 Preliminaries **3**
2.1 Configurations and the disjointness graph . . . 3
2.2 Two unconditional bounds . . . 3
2.3 Orientation data . . . 4

3 Two forbidden patterns **4**
3.1 The small pattern . . . 4
3.2 The larger pattern . . . 4
3.3 How the patterns are proved non-realizable . . . 4

4 The covering theorem **5**
4.1 The $I = 42$ case . . . 5
4.2 The $I = 40$ case . . . 5

*Northeastern University. This work was carried out with substantial generative-AI assistance; see §11.

5 Excluding 40 and 42 **5**

6 The lower bound **5**

7 From orientation signs to the plane **6**
7.1 Integer coordinates are no restriction 6
7.2 Counting points, not side pairs 6
7.3 Removing the collinearity assumption 6

8 What is imported **6**
8.1 An attempt to remove it 7

9 Machine verification, and its limits **7**
9.1 Independent corroboration 7

10 Related work **7**

11 Status, and what we ask of a reader **8**

Disclosure of generative-AI use **8**

1 Introduction

1.1 The problem

Two simple polygons in the plane, with m and n sides, are in *general position* if no vertex of either lies on the boundary of the other and no three of the combined $m + n$ vertices are collinear. Write $f(m, n)$ for the maximum, over such pairs, of $|\partial P \cap \partial Q|$.

Problem (Dillencourt–Mount–Saalfeld, 1993). Determine $f(m, n)$.

They determined $f(m, n)$ when at least one of m, n is even. The odd–odd cases are open. A 2012 preprint of Günther claiming $f(n, n)$ for odd n was withdrawn by its author, with the stated reason that a counterexample was found to a weakened form of one of its lemmas. The smallest open case is $m = n = 7$.

Two elementary facts frame it. Each side of P meets each side of Q at most once in general position, so $I \leq 49$; and since ∂P and ∂Q are closed curves meeting transversally, I is even. The published state of the art gives $I \leq 40$, and 38 is attained. So the entire question is whether 40 is attainable.

1.2 What is proved here

Theorem 1.1 (Main, conditional)**.** *Let P and Q be simple heptagons in the plane, each simple as a point set, such that no vertex of either lies on the boundary of the other. Then*

$$|\partial P \cap \partial Q| \;\leq\; 38,$$

and 38 *is attained. No condition on collinearity of vertices is assumed.*

The hypothesis "no vertex of either lies on the boundary of the other" is exactly the conclusion of Ackerman–Keszegh–Rote §9. We do not prove it; §8 states precisely what we import and why, and §8.1 reports an unsuccessful attempt to remove it.

Remark 1.2. Theorem 1.1 is **not** $f(7, 7) = 38$, and we are careful not to write it that way anywhere. It is the statement one obtains after the general-position reduction has been applied, and the reduction is imported.

1.3 What is new

The contribution is a reduction, not a technique. Both ingredients we use — encoding realizability of a crossing pattern as an oriented-matroid satisfiability problem, and refuting it mechanically — are standard; see §10. What appears to be new is:

- **Two patterns suffice** (Theorem 4.1). A search that was 13,003,620 cases becomes two non-realizability facts plus a degree argument over 28 cases.
- **The smaller pattern is human-readable** and needs no simplicity hypothesis: it constrains three consecutive sides of each polygon and prescribes eight incidences (§3).
- **The reduction to the plane is unconditional.** The combinatorial argument is about orientation signs; getting from there to genuine point sets in $\mathbb{R}^2$ — including removing the collinearity assumption that AKR §3 asserts without proof — is carried out in full (§7).

A bounded prior-art search found no previous statement of the covering result. We stress in §10 that this is weak evidence and not a novelty claim.

2 Preliminaries

2.1 Configurations and the disjointness graph

Let P have vertices $p_0, \dots, p_6$ and Q have $q_0, \dots, q_6$, indices mod 7, with sides $P_i = [p_i, p_{i+1}]$ and $Q_j = [q_j, q_{j+1}]$. For points a, b, c write

$$\mathrm{or}(a,b,c) = (b_1 - a_1)(c_2 - a_2) - (b_2 - a_2)(c_1 - a_1),$$

twice the signed area. Sides $[a,b]$ and $[c,d]$ *cross* when

$$\mathrm{or}(a,b,c) \cdot \mathrm{or}(a,b,d) < 0 \quad \text{and} \quad \mathrm{or}(c,d,a) \cdot \mathrm{or}(c,d,b) < 0.$$

This is a conjunction of *strict* inequalities, a fact used repeatedly: it makes crossing an open condition.

Definition 2.1. The *disjointness graph* G^D is the bipartite graph on $\{P_0, \dots, P_6\} \sqcup \{Q_0, \dots, Q_6\}$ with an edge P_iQ_j when P_i and Q_j do *not* cross.

In general position each side pair contributes at most one intersection point, and exactly one when it crosses, so writing I for the number of intersection points and $|E^D|$ for the number of edges of G^D,

$$I + |E^D| = 49. \tag{1}$$

2.2 Two unconditional bounds

Lemma 2.2. *I is even.*

Lemma 2.3. *$\delta(G^D) \geq 1$, and consequently $I \leq 42$.*

Proof sketch. Every side of P fails to cross at least one side of Q and vice versa, so $|E^D| \geq 7$; with (1) this gives $I \leq 42$. □

Remark 2.4. Černý, Kynčl and Tóth obtain $\delta(G^D) \geq 1$ from the fact that a line meets a simple 7-gon in at most 6 sides. We do not import that: the minimum-degree statement is derived directly and is machine-checked unconditionally, so their Lemma 1 is corroboration here, not a dependency.

By Lemmas 2.2 and 2.3, $I \in \{\dots, 38, 40, 42\}$, and since 38 is attained (§6) the theorem reduces to excluding 40 and 42.

2.3 Orientation data

All of §§3–5 concerns only the signs of the $\binom{14}{3} = 364$ determinants $\mathrm{or}(v_a, v_b, v_c)$ on the combined vertex set. A *configuration* is a 14-tuple of points together with the requirement that all 364 signs be nonzero; a *crossing pattern* is the resulting 7×7 Boolean matrix of side-pair crossings. Realizability questions about patterns are therefore questions about which sign vectors arise from actual point sets — that is, about chirotopes.

3 Two forbidden patterns

3.1 The small pattern

Definition 3.1. Pattern Π_8 constrains sides P_0, P_1, P_2 and Q_0, Q_1, Q_2:

	Q_0	Q_1	Q_2
P_0	$\times$	$\times$	$\times$
P_1	$\times$	$\cdot$	$\times$
P_2	$\times$	$\times$	$*$

where $\times$ means *crosses*, $\cdot$ means *does not cross*, and $*$ is unconstrained. That is eight prescribed incidences: seven crossings and one non-crossing.

Theorem 3.2. *No configuration realizes* Π_8.

Theorem 3.2 assumes nothing beyond the eight incidences and the nondegeneracy of the 364 determinants: in particular it needs *no* hypothesis that P or Q is simple. It is the single most compact statement in this work, and we suggest it as the first thing a sceptical reader should check.

3.2 The larger pattern

Definition 3.3. Pattern Π_{14} constrains $P_0, \ldots, P_3$ and $Q_0, \ldots, Q_4$ by fourteen incidences,

	Q_0	Q_1	Q_2	Q_3	Q_4
P_0	$*$	$\times$	$\times$	$*$	$\times$
P_1	$\times$	$\cdot$	$\times$	$\times$	$\times$
P_2	$\times$	$\times$	$*$	$*$	$\times$
P_3	$\times$	$\times$	$\times$	$*$	$*$

together with seven simplicity conditions: $P_0 \not\times P_2$, $P_0 \not\times P_3$, $P_1 \not\times P_3$, $Q_0 \not\times Q_2$, $Q_0 \not\times Q_3$, $Q_1 \not\times Q_3$, $Q_1 \not\times Q_4$.

Theorem 3.4. *No configuration realizes* Π_{14}.

Unlike Π_8, this one genuinely uses that the polygons do not cross themselves; the seven self-non-crossing conditions are part of the hypothesis and are supplied by simplicity.

3.3 How the patterns are proved non-realizable

Both theorems are established by encoding realizability as a Boolean satisfiability problem over the 364 orientation variables — Grassmann–Plücker relations, acyclicity, and the prescribed incidences — and refuting it. The refutation is not trusted:

1. Each clause of the encoding is *derived from the geometry* inside the proof assistant. The Grassmann–Plücker and acyclicity clauses are polynomial identities in the coordinates; the pattern clauses are the definition of crossing unfolded. Nothing about the encoding is assumed.

2. The solver's refutation is checked, then transcribed into an explicit chain of unit-propagation steps. The kernel verifies a chain of implications between propositions; *no search runs inside the proof assistant.*

Remark 3.5 (Positive control). A refutation that also covered a configuration which exists would be proving something false. Both patterns require P_1 to miss Q_1; in the 38-crossing configuration of §6 those two sides cross, so neither pattern applies to it. This control is itself machine-checked.

4 The covering theorem

Theorem 4.1. *Every crossing pattern of two simple heptagons with $I = 40$, and every one with $I = 42$, contains Π_8 or Π_{14} after relabelling.*

Here "after relabelling" refers to the group of order 392 generated by rotating and reflecting each polygon and exchanging P with Q.

4.1 The $I = 42$ case

By (1), $I = 42$ forces $|E^D| = 7$, and with $\delta(G^D) \geq 1$ on both sides the graph is a perfect matching. There are $7! = 5040$ of these, all equivalent under relabelling to few orbits, and each contains one of the two patterns.

4.2 The $I = 40$ case

Here $|E^D| = 9$, so G^D has nine edges with all fourteen degrees at least 1. Write $r_i = \deg P_i$ and $c_j = \deg Q_j$, so $\sum_i r_i = \sum_j c_j = 9$ with every $r_i, c_j \geq 1$. Since nine edges are spread over seven rows each of degree at least one, the multiset $\{r_i\}$ is one of few shapes, and likewise for columns; enumerating gives **28 degree shapes**. For each shape a short argument locates a sub-pattern equivalent to Π_8 or Π_{14}.

The point of organising the argument this way is that the 28 cases are a human-sized object, whereas the 13,003,620 patterns are not.

5 Excluding 40 and 42

Corollary 5.1. *No configuration has $I = 40$ or $I = 42$.*

Proof. Immediate from Theorems 3.2, 3.4 and 4.1. □

Corollary 5.2. *Every configuration has $I \leq 38$.*

Proof. $I \leq 42$ by Lemma 2.3, I is even by Lemma 2.2, and 40 and 42 are excluded by Corollary 5.1. □

6 The lower bound

Proposition 6.1. *The heptagons with vertices*

$$P : (373, 157),\ (132, 5),\ (369, 182),\ (88, 157),\ (327, 282),\ (60, 272),\ (372, 298),$$
$$Q : (344, 24),\ (89, 303),\ (345, 19),\ (373, 214),\ (270, 338),\ (325, 93),\ (192, 284)$$

are simple, are in general position, and their boundaries meet in exactly 38 points.

All three assertions are decided by exact integer arithmetic and verified by the kernel. Combined with Corollary 5.2 this gives the equality in Theorem 1.1.

7 From orientation signs to the plane

Sections 3–5 are about sign vectors. Three further steps are needed to reach Theorem 1.1, and all three are carried out unconditionally.

7.1 Integer coordinates are no restriction

The combinatorial development uses integer coordinates. This costs nothing: or is homogeneous of degree 2, so the set of point-tuples with a given sign vector is invariant under scaling. Scaling a configuration until a box around it has side length exceeding 1 produces an integer point in that box with the same 364 signs. No rationals and no clearing of denominators are required.

7.2 Counting points, not side pairs

The combinatorial argument counts crossing *pairs*; the theorem counts *points*. In general position these agree: a crossing pair meets in exactly one point, and a point determines its pair. Both directions are proved.

7.3 Removing the collinearity assumption

AKR §3 assumes, "for convenience" and without proof, that no three of the combined vertices are collinear. We do not assume it. The argument is a perturbation, in two halves:

Lemma 7.1. *Crossing is an open condition, so for a small enough perturbation of the fourteen vertices no existing crossing is lost.*

Lemma 7.2. *If two segments are disjoint, they remain disjoint when their four endpoints move by less than some $\varepsilon > 0$.*

Lemma 7.1 handles the crossings; Lemma 7.2 handles simplicity, which is a *closed* condition and therefore out of reach of any openness argument. Since the configurations in general position are dense, combining the two lemmas approximates an arbitrary pair of point-set simple heptagons by one in general position with at least as many crossings, giving the bound for the original. Lemma 7.2 is proved for an arbitrary real normed space and is independent of everything else here.

Remark 7.3. Note the direction: we need the perturbed configuration to have *at least* as many crossings, not the same number, and openness gives exactly that.

8 What is imported

Hypothesis 8.1 ((E9))**.** No vertex of either polygon lies on the boundary of the other.

This is the conclusion of AKR §9, where §9.1 removes vertex–vertex coincidences and §9.2 vertex–side ones. It is the *only* external mathematical input to Theorem 1.1.

Why it is needed, precisely: without it, a point of $\partial P \cap \partial Q$ can lie on *no* crossing pair. Two sides that merely touch share a point but have vanishing determinants, so the pair is not counted, and the bound on pairs does not bound points. The failure is in the unfavourable direction, which is why §7.3 does not also dispose of it.

Remark 8.2 (Citation). It would be an under-citation to attribute all of the usual general-position package to §9. §9 gives Hypothesis 8.1 and nothing more; the collinearity clause has its only warrant in one unproved sentence of §3, and is discharged here (§7.3). Additionally, that the perturbed polygons remain simple is stated nowhere in that paper. We fold simplicity into Hypothesis 8.1 deliberately, so that the gap is visible.

8.1 An attempt to remove it

We attempted to prove Hypothesis 8.1 and did not succeed. The decomposition into its vertex–vertex and vertex–side halves is formalized, as are two local lemmas, but the global reduction is not. Two negative observations came out of the attempt and are worth recording, because they close off the obvious routes:

- Generic perturbation preserving proper crossings does *not* imply that the point count is monotone.
- Naive displacement into the interior of a cone does not either.

Explicit coordinates exhibiting both failures are recorded in the project repository; they are analytic and *not* machine-checked. This is consistent with the known bad case $\mathrm{Cone}_P(v) \subseteq \mathrm{Cone}_Q(w)$, on which an earlier attempt of ours broke. None of this is evidence that AKR's theorem is false, and we allege no error in it; what remains is substantial unfinished work.

9 Machine verification, and its limits

Every mathematical assertion in §§2–7 is verified by the Lean 4 kernel. There is no `sorry`, no `native_decide`, and no project-specific axiom; every theorem depends on at most the three standard axioms `propext`, `Classical.choice`, `Quot.sound`. Hypothesis 8.1 is a hypothesis of the statement, never an axiom.

Remark 9.1 (What the kernel does not check)*.* The kernel proves theorems about definitions; it does not certify that the definitions mean what we say. That $\partial P \cap \partial Q$ is modelled by the right set, and that "simple heptagon" is modelled by the right predicate, are judgements a human reader must make. This is, in our view, the principal remaining risk, and it is not reducible by more computation.

One partial check is available and we have carried it out: the definitions, applied to the explicit 38-crossing configuration of Proposition 6.1, yield exactly 38. A wrong definition would have to be wrong in a way that still lands on that value.

9.1 Independent corroboration

Beyond the kernel: the terminal case analysis of an earlier version of this argument was reproduced by a second, independently written implementation in a different language; all solver refutations are re-checked by a standalone proof checker, which is itself first shown to reject non-refutations; and all generated artifacts are regenerated and compared byte-for-byte. These guard against implementation error. *None of them is a check on whether the argument is the right argument.*

10 Related work

Dillencourt, Mount and Saalfeld posed the problem and settled the even cases. Černý, Kynčl and Tóth gave bounds for odd cases and solved $k = 5$, explicitly leaving $k = \ell = 7$ open. Ackerman, Keszegh and Rote obtained asymptotically optimal bounds; their §9 is what we import. Günther's 2012 preprint on odd n was withdrawn by its author.

Both of our tools are standard. Encoding order-type or chirotope realizability as a satisfiability problem and refuting it mechanically is well established, including in combination with proof assistants. Framing extremal results in terms of forbidden configurations is likewise standard. What we have not found in the literature is either applied to crossing patterns of two simple polygons, or to $f(m, n)$.

Remark 10.1 (On novelty). We searched for prior statements of the covering result and found none. We record plainly that this is **weak evidence and not a novelty claim**: the search was bounded, the journal version of one central reference was not reachable, the forward citations of another were not examined, and the relevant literature is small enough that a result could sit in it under different vocabulary. A domain expert's judgement is not replaced by such a search.

11 Status, and what we ask of a reader

To state the position exactly:

- $f(7,7) = 38$ is **not** proved. Theorem 1.1 carries Hypothesis 8.1.
- Everything else is machine-checked from primitive definitions with no published lemma imported.
- **No mathematician outside this process has reviewed any of it.**

The third point is the binding constraint, and it is why this paper exists. Two entry points need the least context:

1. **Theorem 3.2.** Eight incidences on three consecutive sides of each polygon, no simplicity hypothesis. If it is wrong, everything here is wrong.
2. **Lemma 7.2.** A self-contained statement about segments in a normed space, whose formalization depends on nothing else in the development.

We would also welcome an opinion on Remark 9.1 — whether our formal definitions capture the intended objects — and on whether the covering result is in fact already known.

Disclosure of generative-AI use

Generative-AI systems were used substantially throughout: exploring the reduction, generating essentially all software including the formalization, auditing the cited sources, and drafting this paper. The named author is responsible for the claims. AI output was not treated as trustworthy: each major claim is discharged by a separately recorded mechanism — source audit, independent reimplementation, a machine-checkable certificate with a standalone checker, or kernel verification — and errors introduced during development are catalogued in the project repository rather than quietly repaired. The patterns of §3 were *found* by computer search; what they are is small enough to check by hand.